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SDC205

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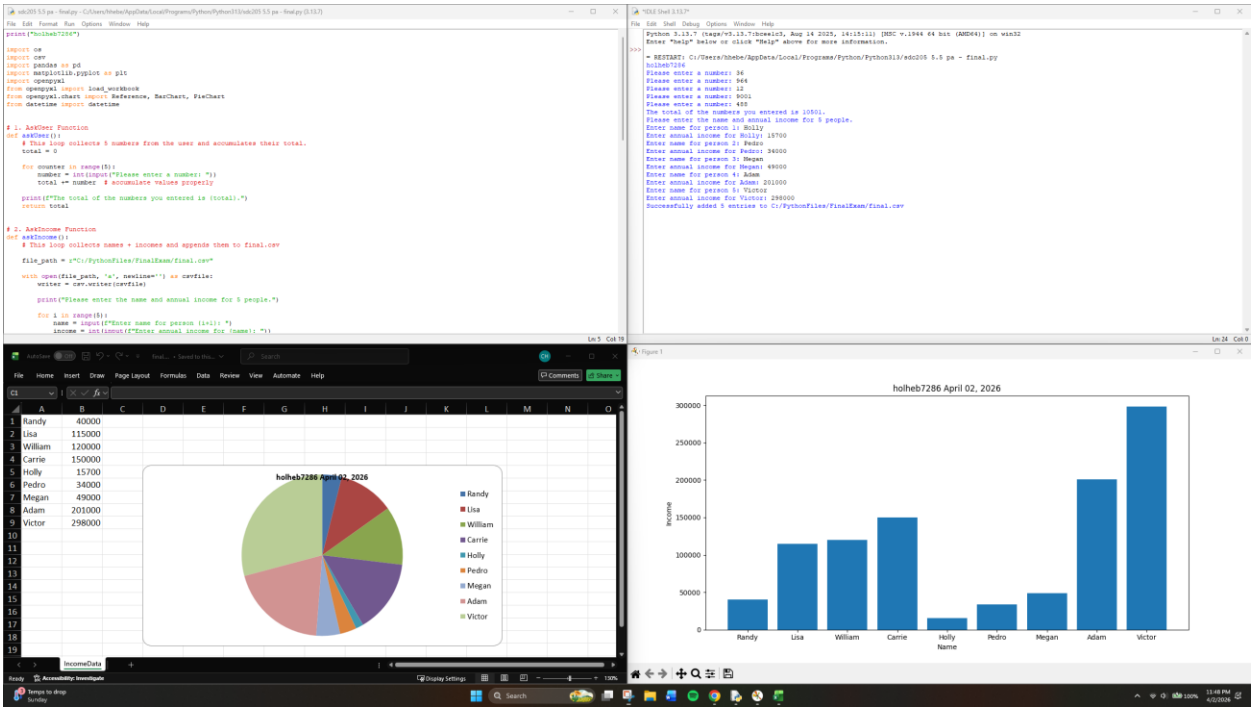
April 2, 2026

Final Performance Assessment

SDC205



sdc205 5.5 pa - final.py - C:/Users/hhebe/AppData/Local/Programs/Python/Python313/sdc205 5.5 pa - final.py (3.13.7)

File Edit Format Run Options Window Help

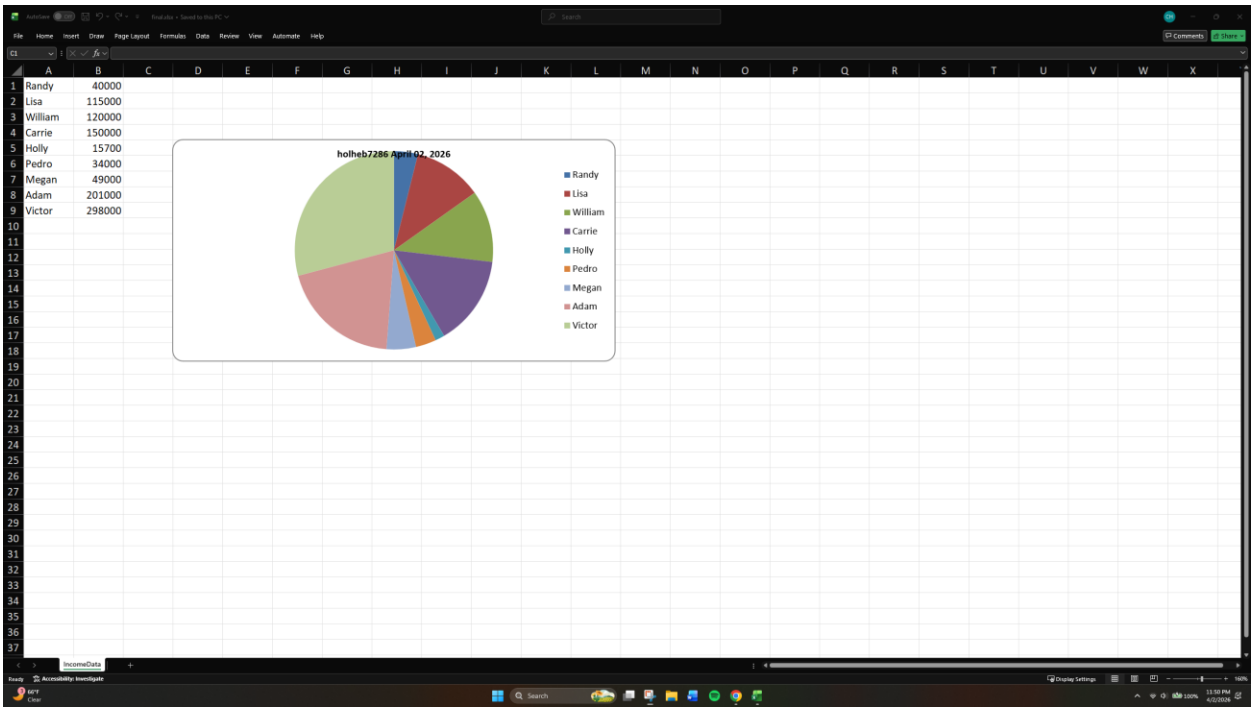
```python
print("holheb7286")

import os
import csv
import pandas as pd
import matplotlib.pyplot as plt
import openpyxl
from openpyxl import load_workbook
from openpyxl.chart import Reference, BarChart, PieChart
from datetime import datetime


# 1. AskUser Function
def askUser():
    # This loop collects 5 numbers from the user and accumulates their total.
    total = 0

    for counter in range(5):
        number = int(input("Please enter a number: "))
        total += number  # accumulate values properly

    print(f"The total of the numbers you entered is {total}.")
    return total


# 2. AskIncome Function
def askIncome():
    # This loop collects names + incomes and appends them to final.csv

    file_path = r"C:/PythonFiles/FinalExam/final.csv"

    with open(file_path, 'a', newline='') as csvfile:
        writer = csv.writer(csvfile)

        print("Please enter the name and annual income for 5 people.")

        for i in range(5):
            name = input(f"Enter name for person {i+1}: ")
            income = int(input(f"Enter annual income for {name}: "))

            # append each entry as a new row
            writer.writerow([name, income])

    print(f"Successfully added 5 entries to {file_path}")


# 3. Excel Pie Chart Function
def excelFile():
    # Load existing CSV into Excel-compatible format
    file_path = r"C:/PythonFiles/FinalExam/final.csv"
    wb = openpyxl.Workbook()
    ws = wb.active
    ws.title = "IncomeData"

    # Read CSV and write into workbook
    with open(file_path, 'r') as f:
        reader = csv.reader(f)
        for row in reader:
            ws.append(row)

    # Convert income column to ints (VERY IMPORTANT for charts)
    for row in range(2, ws.max_row + 1):
        ws[f"B{row}"] = int(ws[f"B{row}"].value)

    # Labels are in Column A (1), Data in Column B (2)
    labels = Reference(ws, min_col=1, min_row=2, max_row=ws.max_row)
    data = Reference(ws, min_col=2, min_row=1, max_row=ws.max_row)

    # Create pie chart
    chart = PieChart()
    chart.title = f"holheb7286 {datetime.now().strftime('%B %d, %Y')}"

    chart.add_data(data)
    chart.set_categories(labels)

    # Place chart in sheet
    ws.add_chart(chart, "D5")

    # Save file
    wb.save(r"C:/PythonFiles/FinalExam/final.xlsx")

# 4. Matplotlib Bar Chart Function
```

Ln: 5 Col: 19

IDLE Shell 3.13.7

File Edit Shell Debug Options Window Help

```
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/hhebe/AppData/Local/Programs/Python/Python313/sdc205 5.5 pa - final.py
holheb7286
Please enter a number: 36
Please enter a number: 964
Please enter a number: 12
Please enter a number: 9001
Please enter a number: 488
The total of the numbers you entered is 10501.
Please enter the name and annual income for 5 people.
Enter name for person 1: Holly
Enter annual income for Holly: 15700
Enter name for person 2: Pedro
Enter annual income for Pedro: 34000
Enter name for person 3: Megan
Enter annual income for Megan: 49000
Enter name for person 4: Adam
Enter annual income for Adam: 201000
Enter name for person 5: Victor
Enter annual income for Victor: 298000
Successfully added 5 entries to C:/PythonFiles/FinalExam/final.csv
```



1. What does implicitly typed mean?

It means you do not have to explicitly declare a variable data type (like integer, text, or decimal) before using it. Python automatically figures out the type based on the value you assign to it (e.g., typing x = 10 automatically makes x an integer).

2. What is code reuse and how do functions facilitate it?

Code reuse is the practice of writing a helpful piece of code once and using it multiple times. Functions facilitate this by letting you "package" a block of code under a specific name. Whenever you need that logic, you just call the function name instead of rewriting all the code.

3. When opening files in Python, why do we specify permissions (modes)?

Specifying permissions (like 'r' for read, 'w' for write, or 'a' for append) tells the operating system exactly what your program is allowed to do. This prevents accidental data destruction, like accidentally overwriting your entire file when you only meant to read it.

4. What is the difference between a list and a dictionary in Python?

- **Lists** are ordered collections of items accessed by their numerical position (e.g., item 0,

item 1, item 2).

- **Dictionaries** are collections of key-value pairs accessed by a unique name or "key" instead of a number (e.g., looking up {'Name': 'Alice'}).

5. What index value do all structures in Python begin at?

Python uses 0-based indexing, meaning all sequences (like lists and strings) start at the index of **0**.

6. What are the three common graph types used in real-world data analytics?

Based on standard data analytics (and the tools we used), the top three are:

- **Bar Graphs** (for comparing categorical data)
- **Pie Charts** (for showing parts of a whole)
- **Line Graphs** (for showing trends over time)

7. What are the two required parameters for the file open() function?

To safely and predictably interact with files, you provide the **File Path** (the name/location of the file) and the **Mode** (what you want to do with it, like 'r' or 'w').

8. Why is it necessary to convert values into integers before performing arithmetic?

When Python reads data from a text or CSV file, it reads everything as **Strings** (plain text). If you try to add the text "10" and "5", Python mashes the text together to get "105". You must convert them to integers (int()) so the computer knows to perform actual math to get 15.

9. What is the difference between a .csv file and an .xlsx file?

- **CSV** is a plain text format where data is simply separated by commas. It is lightweight, fast, and has no formatting.
- **XLSX** is an Excel file. It is a heavier, complex file that supports multiple sheets, bold text, colors, charts, and live formulas.

10. What is the benefit of using an absolute path instead of a relative path?

An absolute path provides the *exact, complete* GPS coordinates to the file starting from your hard drive root (e.g., C:/PythonFiles/final.csv). It guarantees Python will find the file no

matter what folder your script is currently running from, eliminating the dreaded "File Not Found" error.